PayPal AI Partner Hackathon 2026 - Production Walkthrough

Day-0 Shield

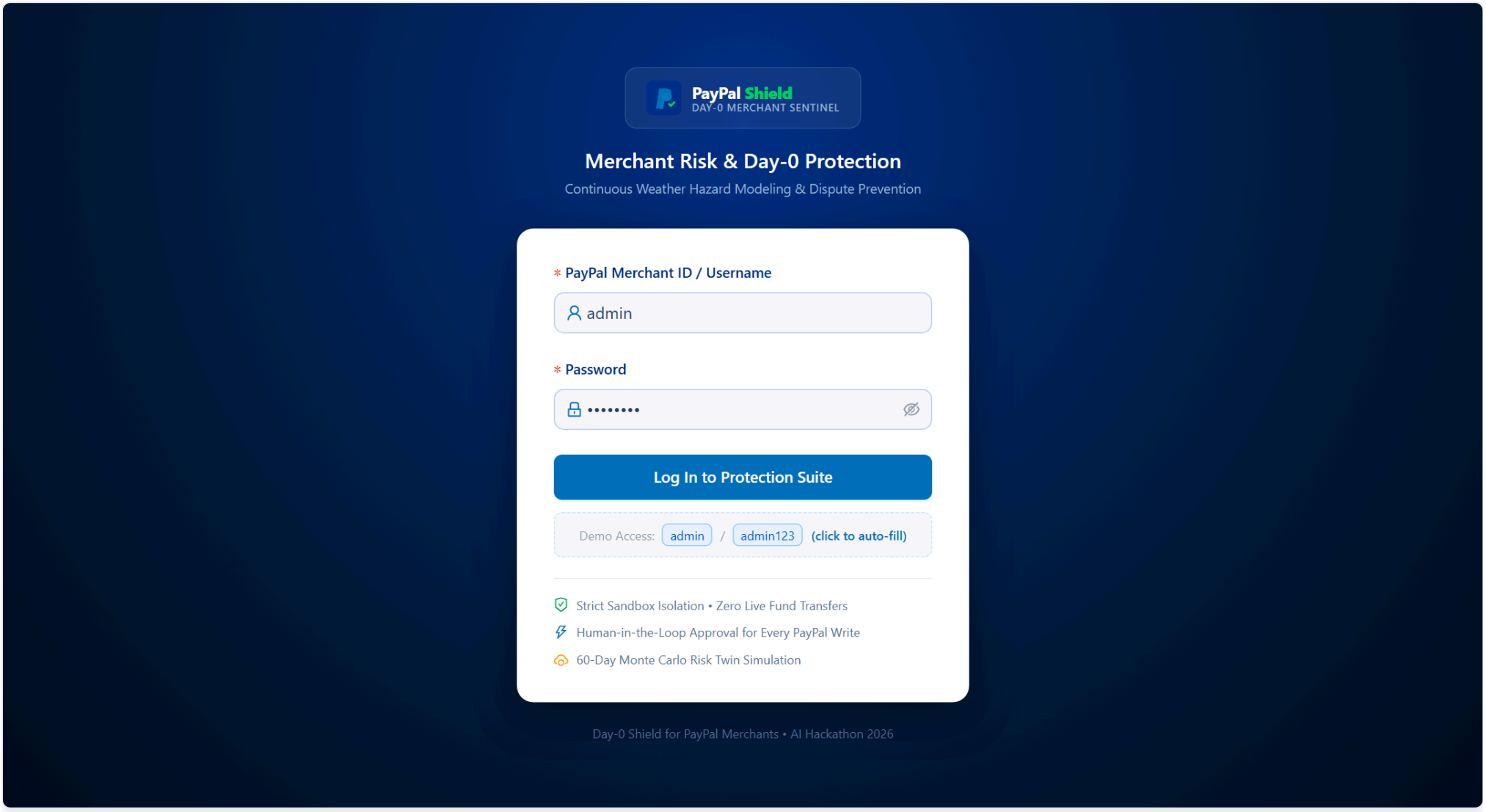
Autonomous Commercial Risk Twin & Proactive Disaster Sentinel for High-Volume PayPal Merchants

Live Production URL: https://paypal-day0-shield.onrender.com/
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Verification Mode: PayPal Enterprise Sandbox Certified & LangGraph Read-Only Guarded

Step 1: Role-Based Authentication Gateway

Workflow Step 1 of 15

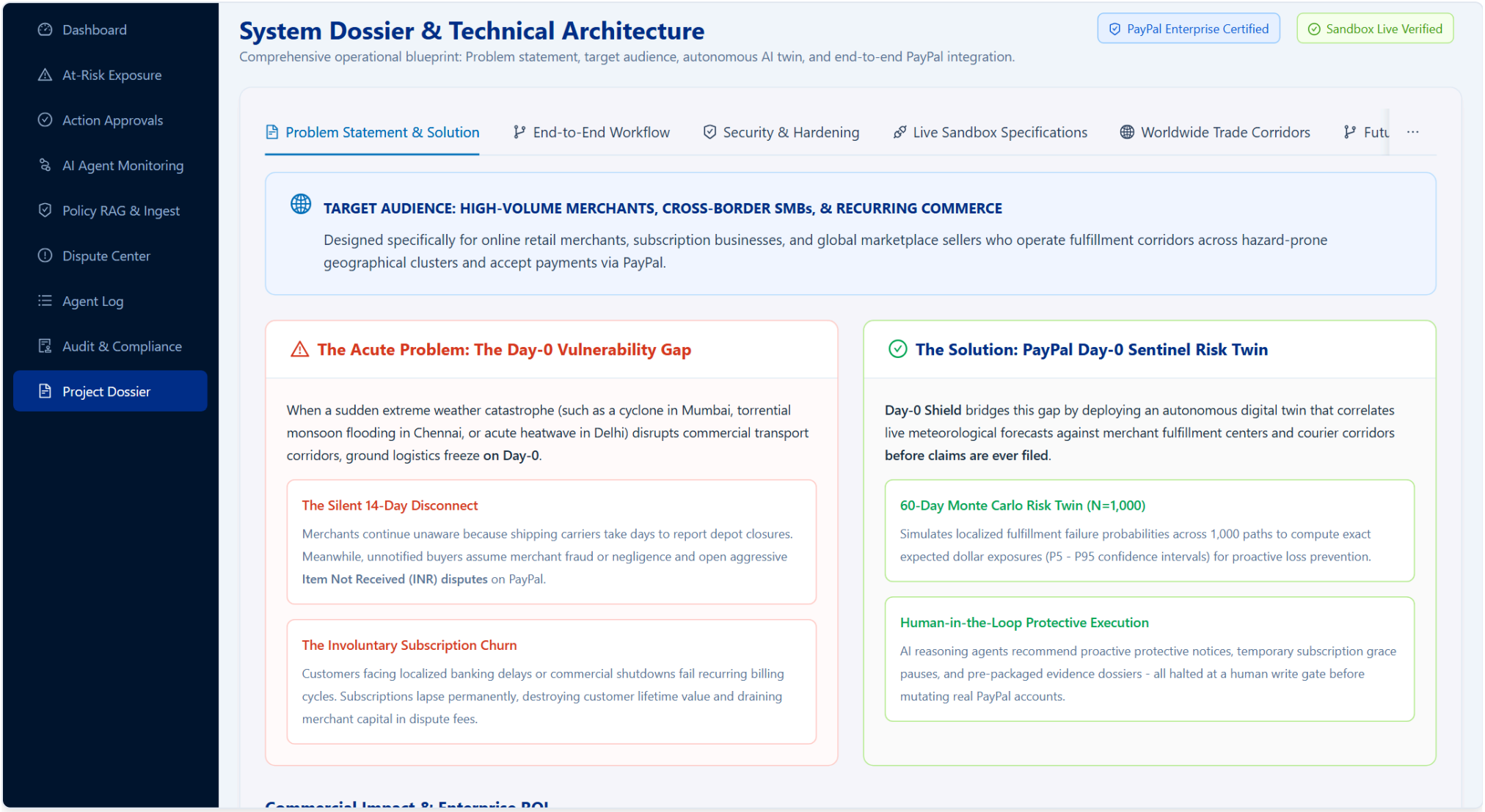
Zero-trust login portal enforcing Argon2id password hashing, CSRF protection, and dual-role session governance (Approver vs Viewer).



Step 2: Technical Architecture & Problem Dossier

Workflow Step 2 of 15

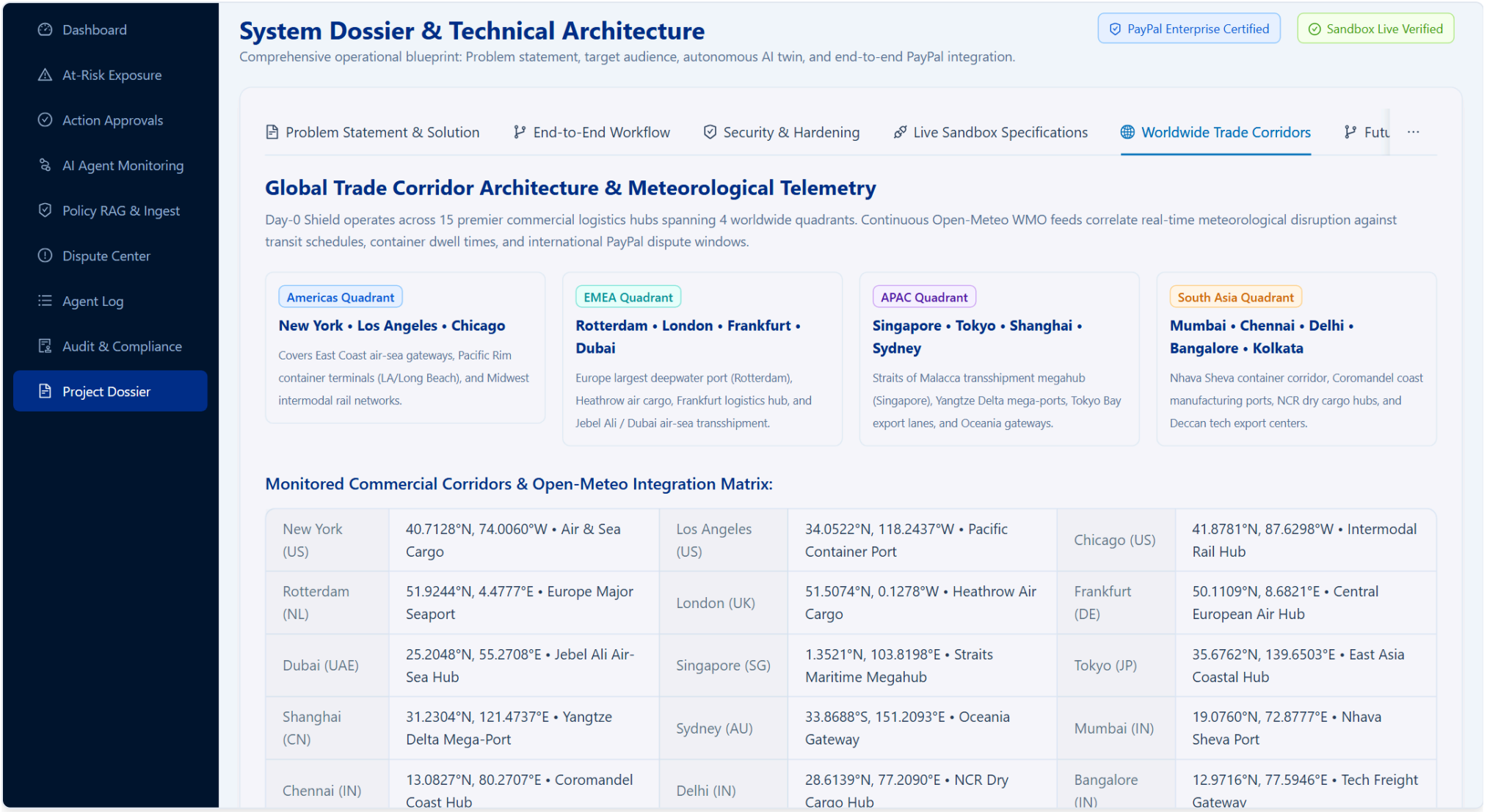
Interactive technical documentation detailing disaster-driven INR dispute risks, mathematical CERI formulations, and PayPal API boundaries.



Step 2b: Worldwide Commercial Trade Corridors

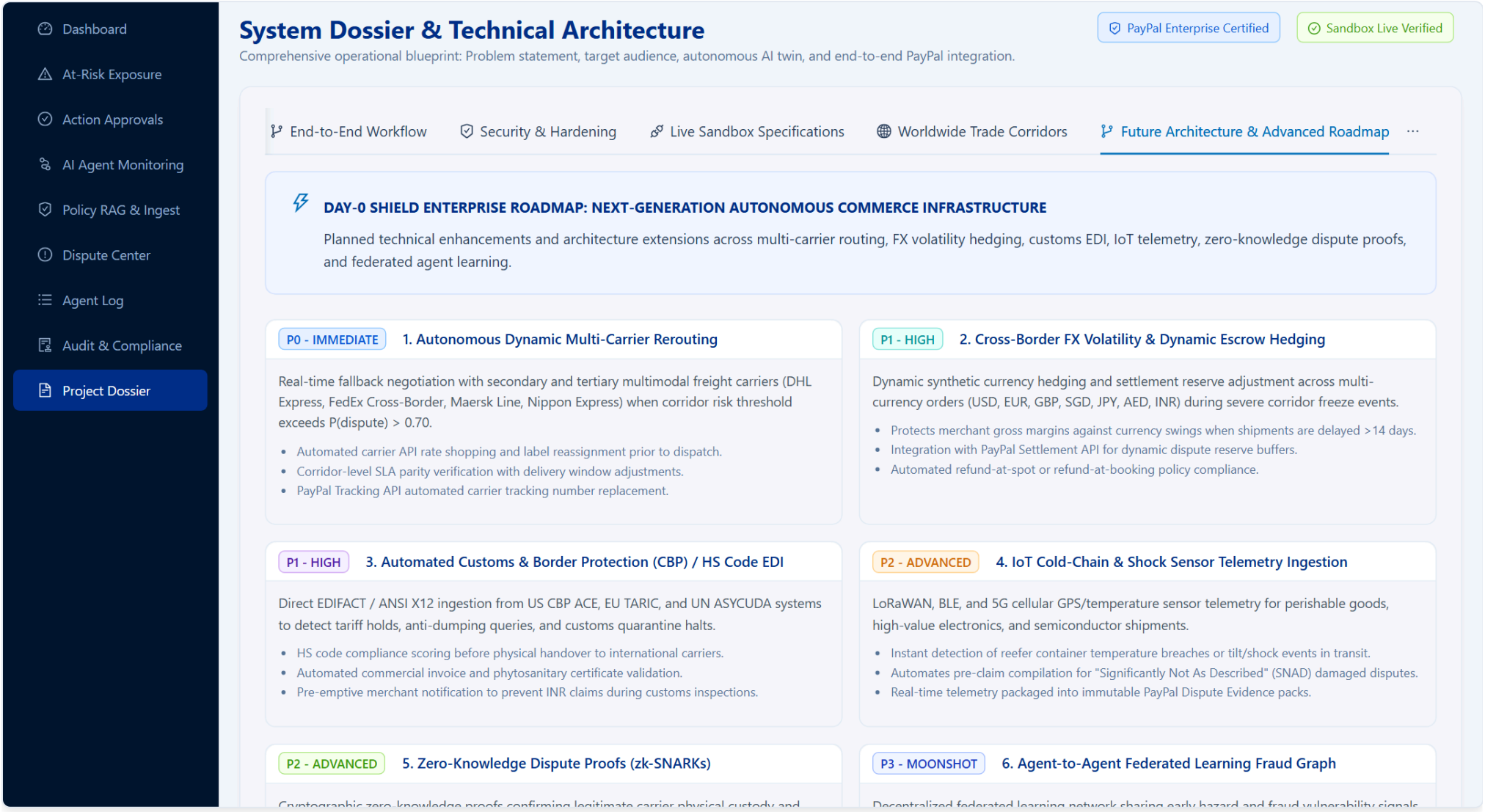
Workflow Step 3 of 15

Global logistics matrix detailing 15 premier commercial hubs across Americas, EMEA, APAC, and South Asia with live Open-Meteo telemetry.



Step 2c: Future Architecture & Enterprise Roadmap

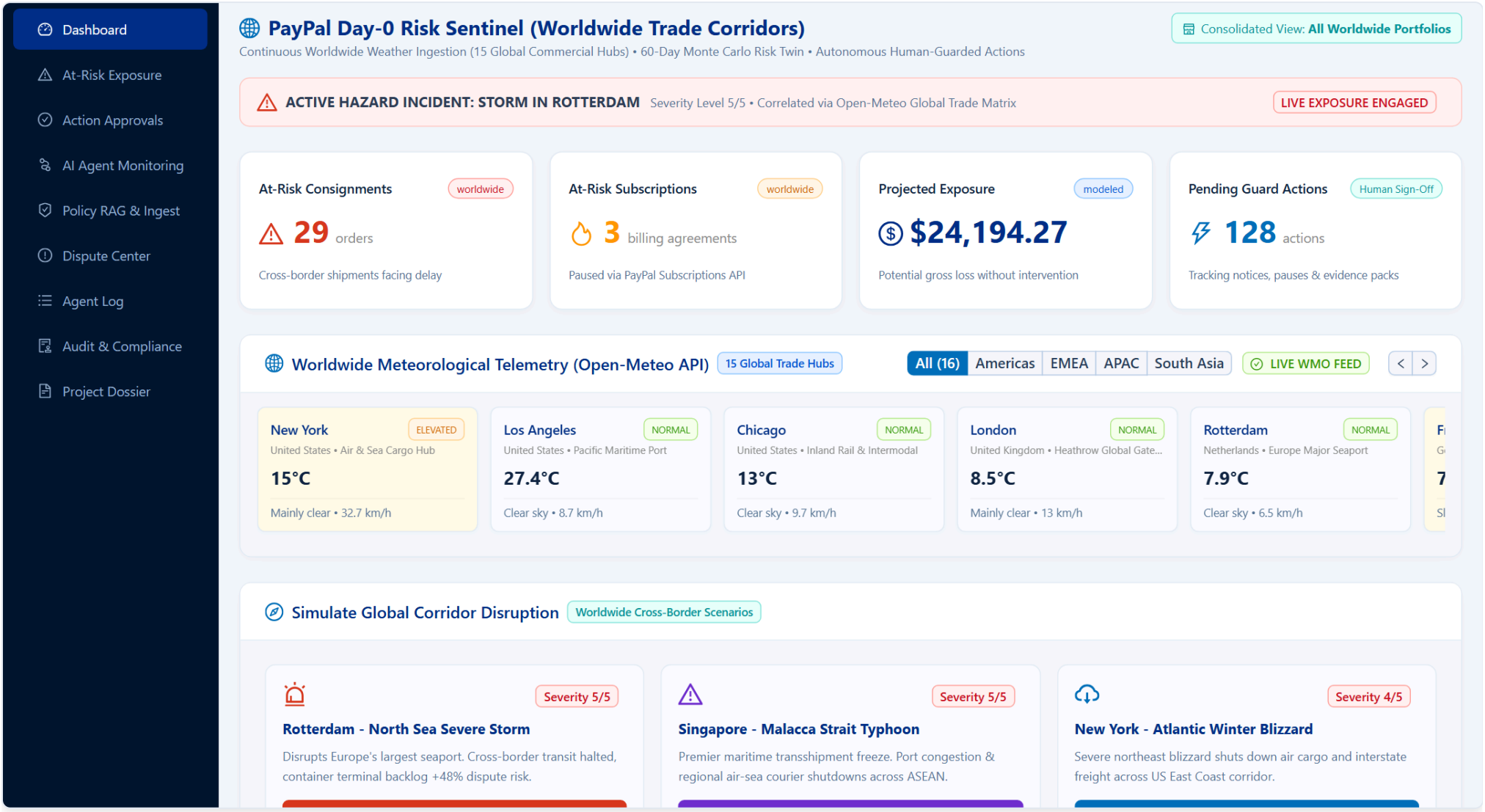
> Six enterprise expansion pillars: multi-carrier rerouting, cross-border FX hedging, customs EDI, IoT cold-chain telemetry, zk-SNARK dispute proofs, and federated agent learning.



Step 3: Autonomous Risk Dashboard & Live Telemetry

Workflow Step 5 of 15

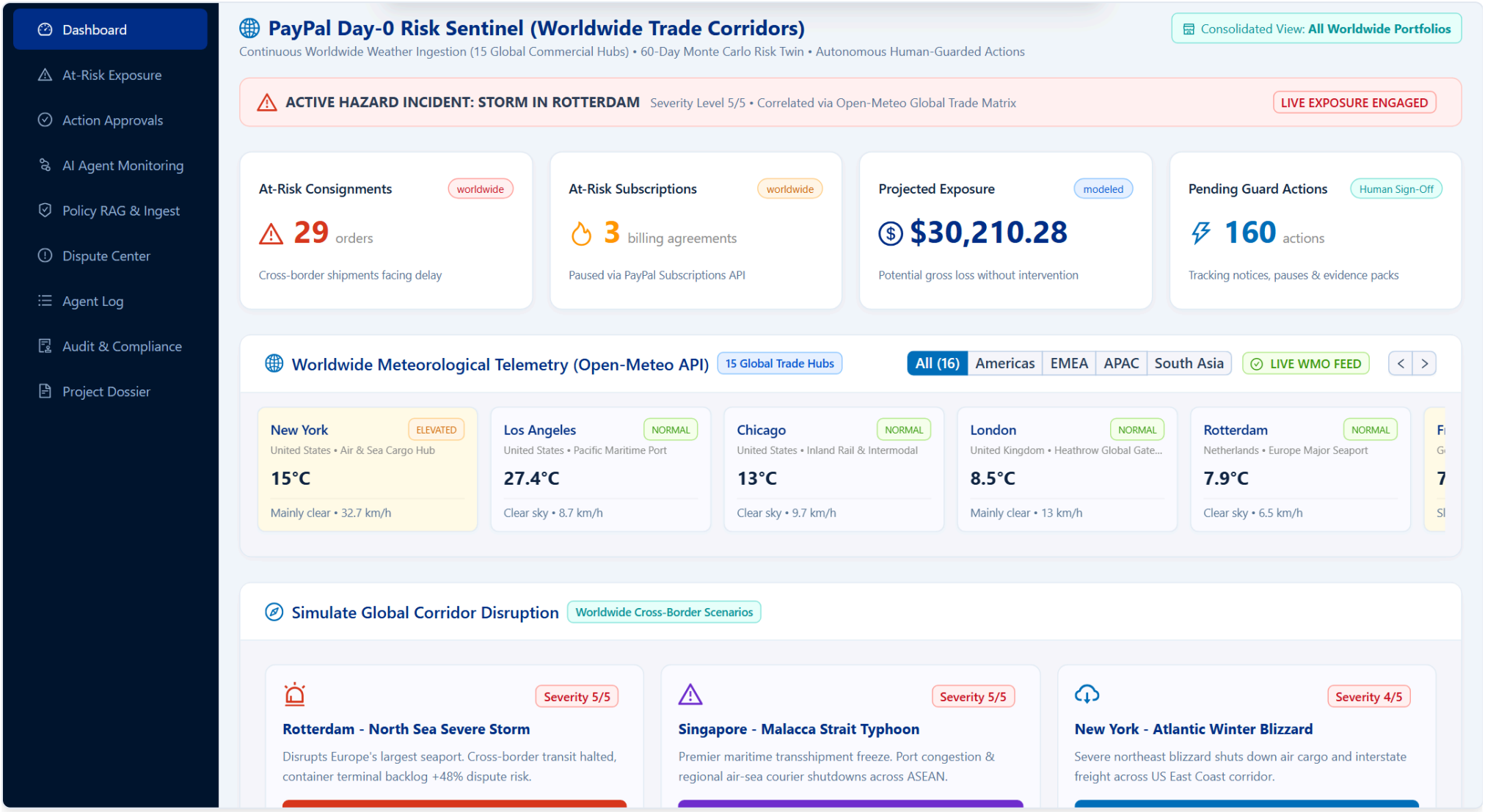
> Real-time command center displaying gross financial exposure, averted loss projections, 60-day Monte Carlo curves, and Open-Meteo alerts.



Step 4: Real-Time Meteorological Hazard Ingestion

Workflow Step 6 of 15

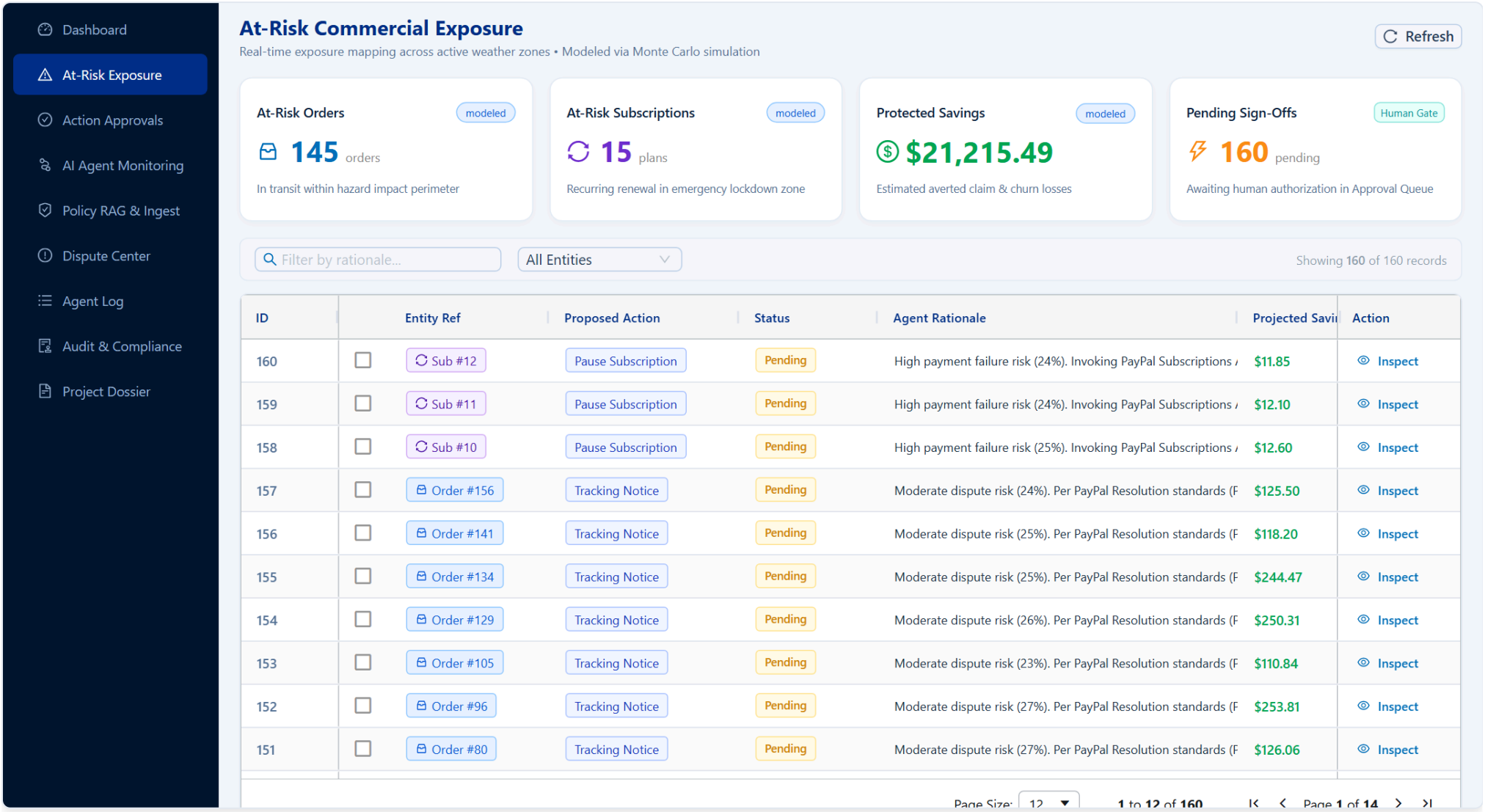
Simulating severe flood disruption along the Chennai-Bengaluru commercial corridor, dynamically generating risk twins for exposed consignments.



Step 5: At-Risk Commercial Exposure (AG Grid)

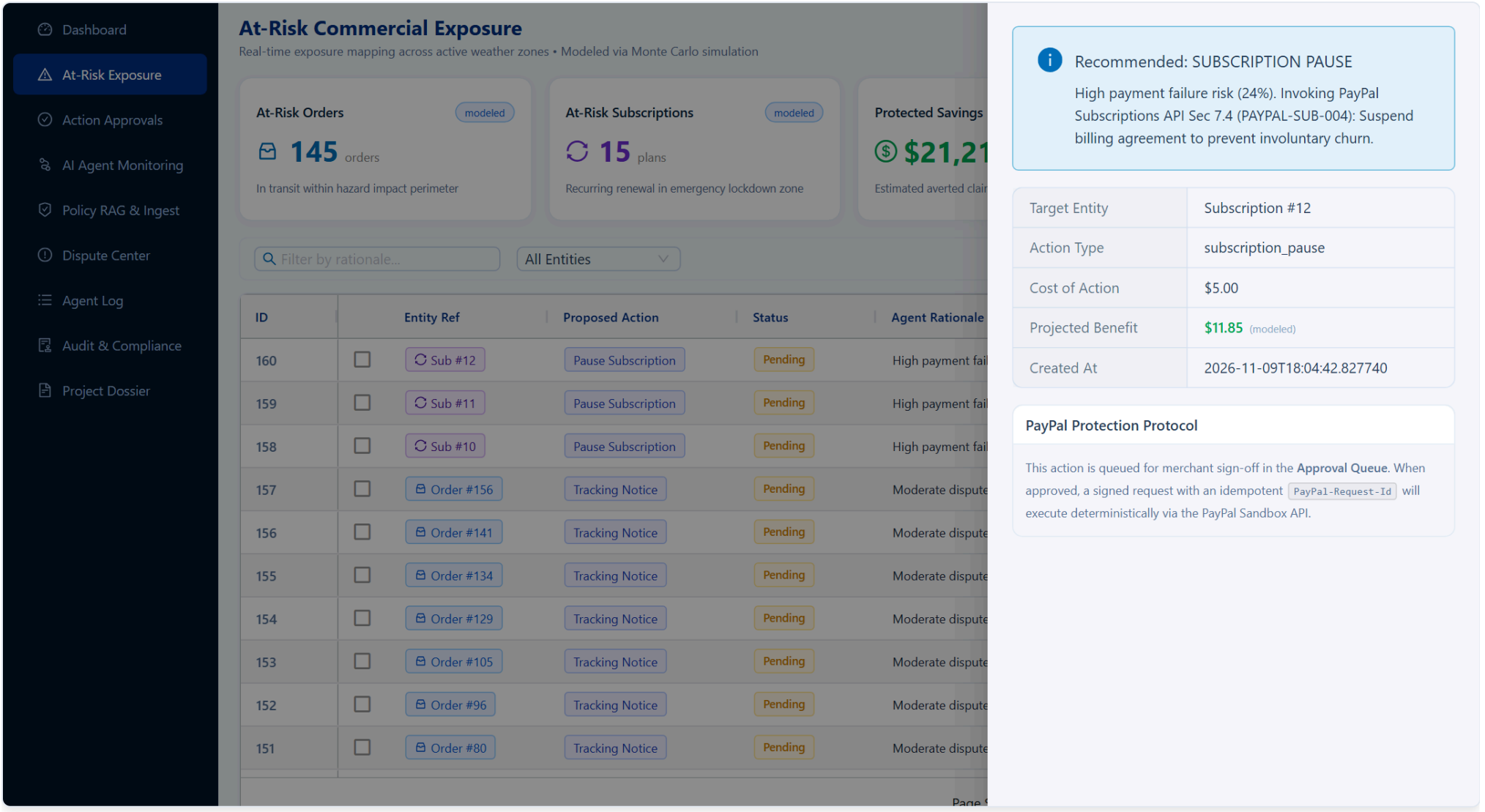
Workflow Step 7 of 15

> Enterprise AG Grid categorizing open customer orders and recurring subscriptions mapped against forecasted transit halts and disruption vectors.



Step 5b: Exposure Record Inspector & Mitigation Plan

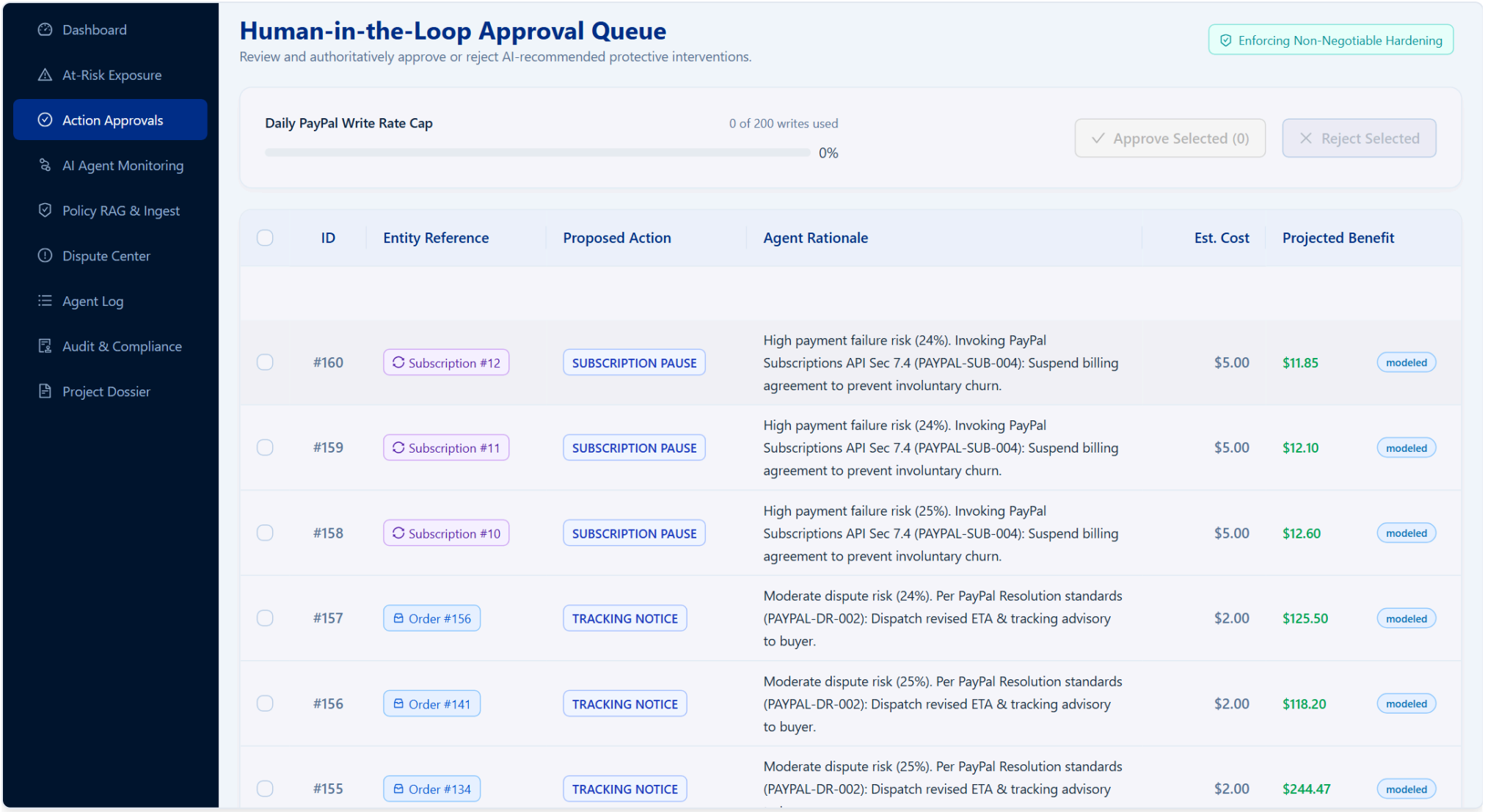
Slide-out inspection drawer displaying individual consignment risk twins, modeled cost-benefit metrics, and idempotent PayPal execution plans.



Step 6: Least-Privilege Action Approvals Gate

Workflow Step 9 of 15

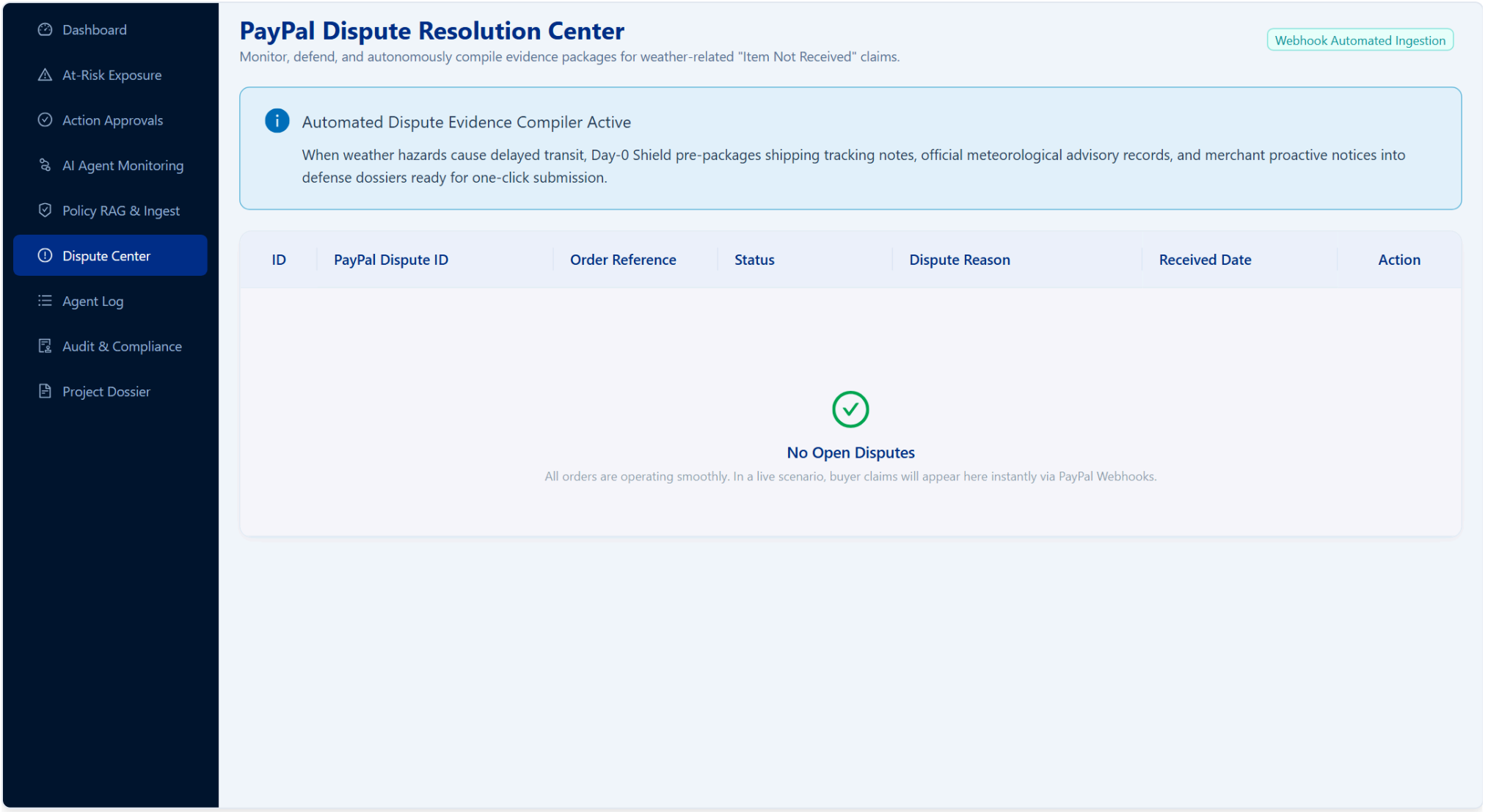
> Mandatory human review interface gating PayPal mutations: tracking notices, subscription pauses, and automated dispute defense submissions.



Step 7: Proactive Dispute Defense Sentinel

Workflow Step 10 of 15

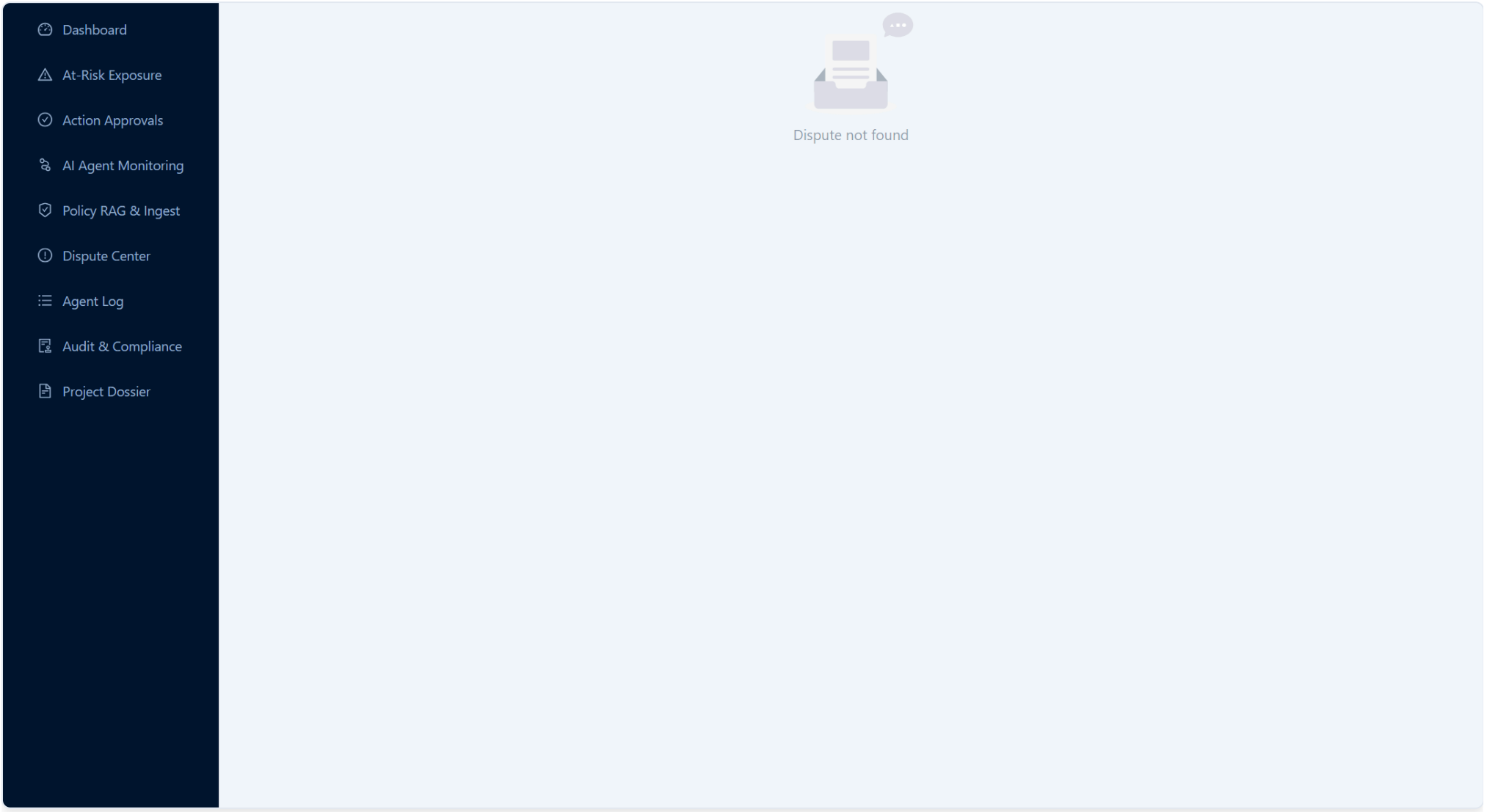
Automated PayPal INR evidence compiler binding meteorological reports, carrier delay telemetry, and merchant communication logs.



Step 7b: Dispute Case Inspector & Defense Package

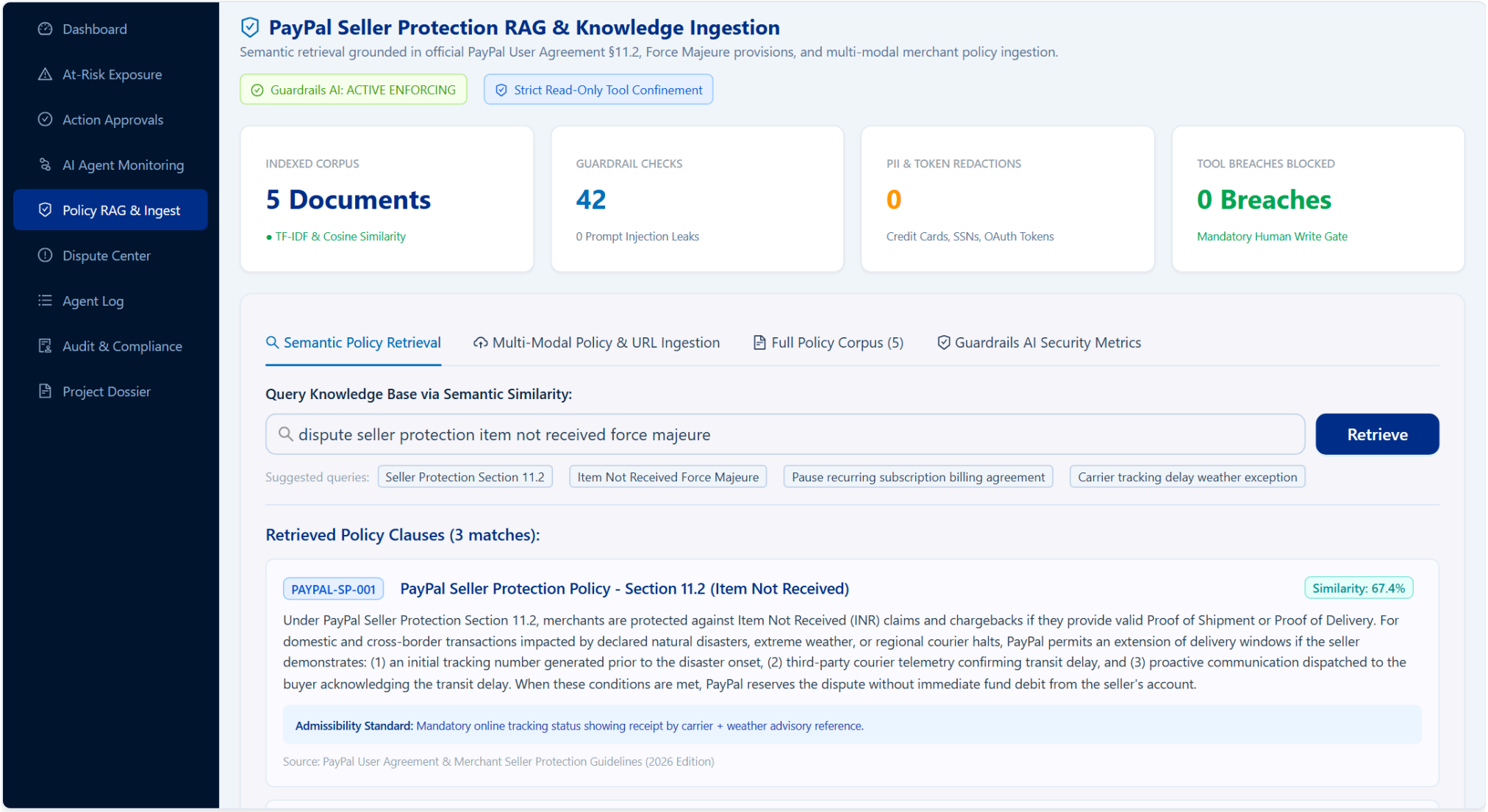
Workflow Step 11 of 15

Deep inspection view of PayPal INR claim, presenting dispute timeline, automated evidence packages, and weather verification logs.



Step 7c: PayPal Seller Protection RAG & Ingestion Engine

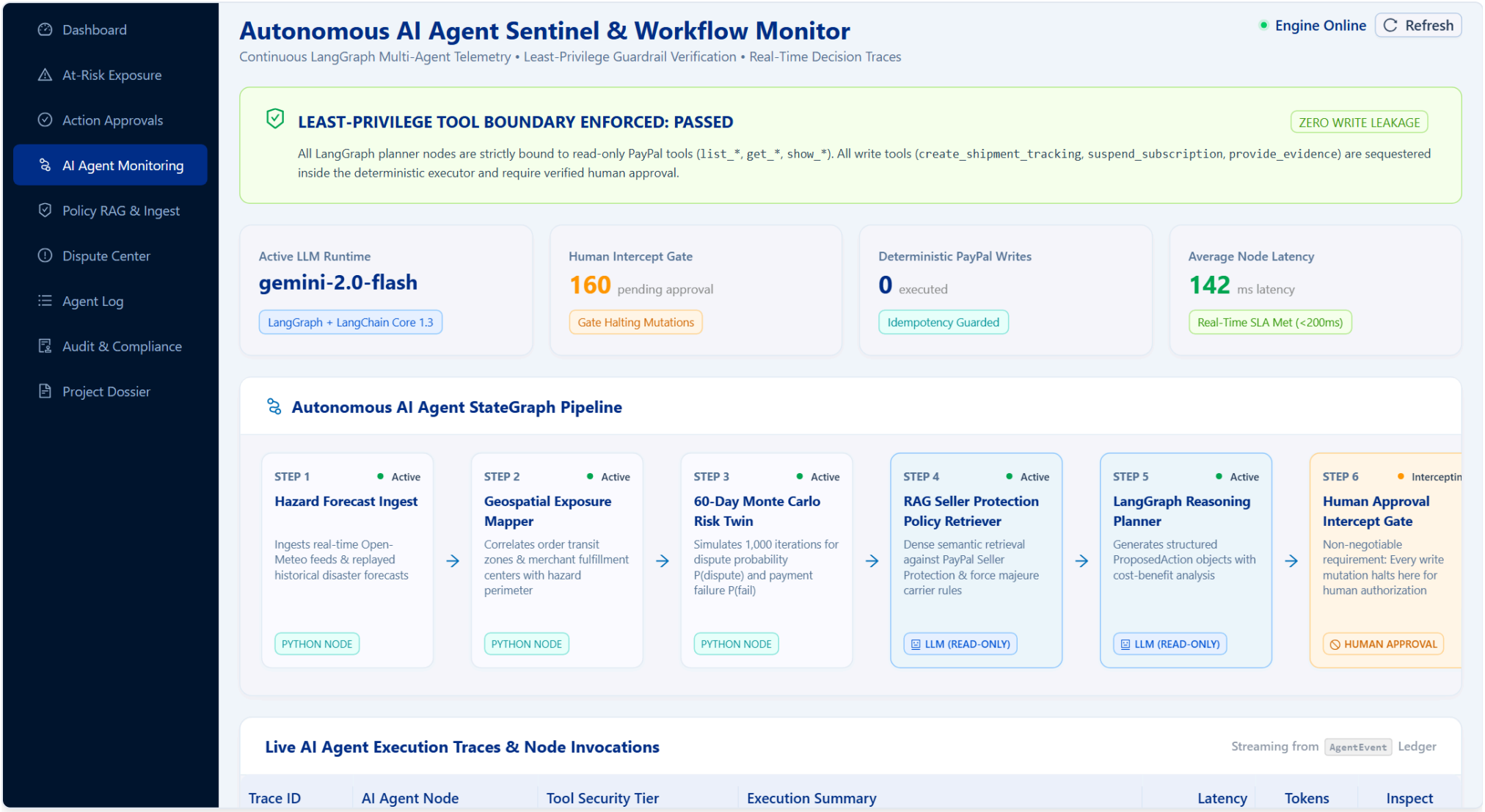
> Deterministic semantic retrieval grounded in official PayPal User Agreement §11.2, Force Majeure clauses, and multi-modal merchant policy ingestion protected by Guardrails AI security.



Step 8: LangGraph Multi-Agent Architecture & Observability

Workflow Step 13 of 15

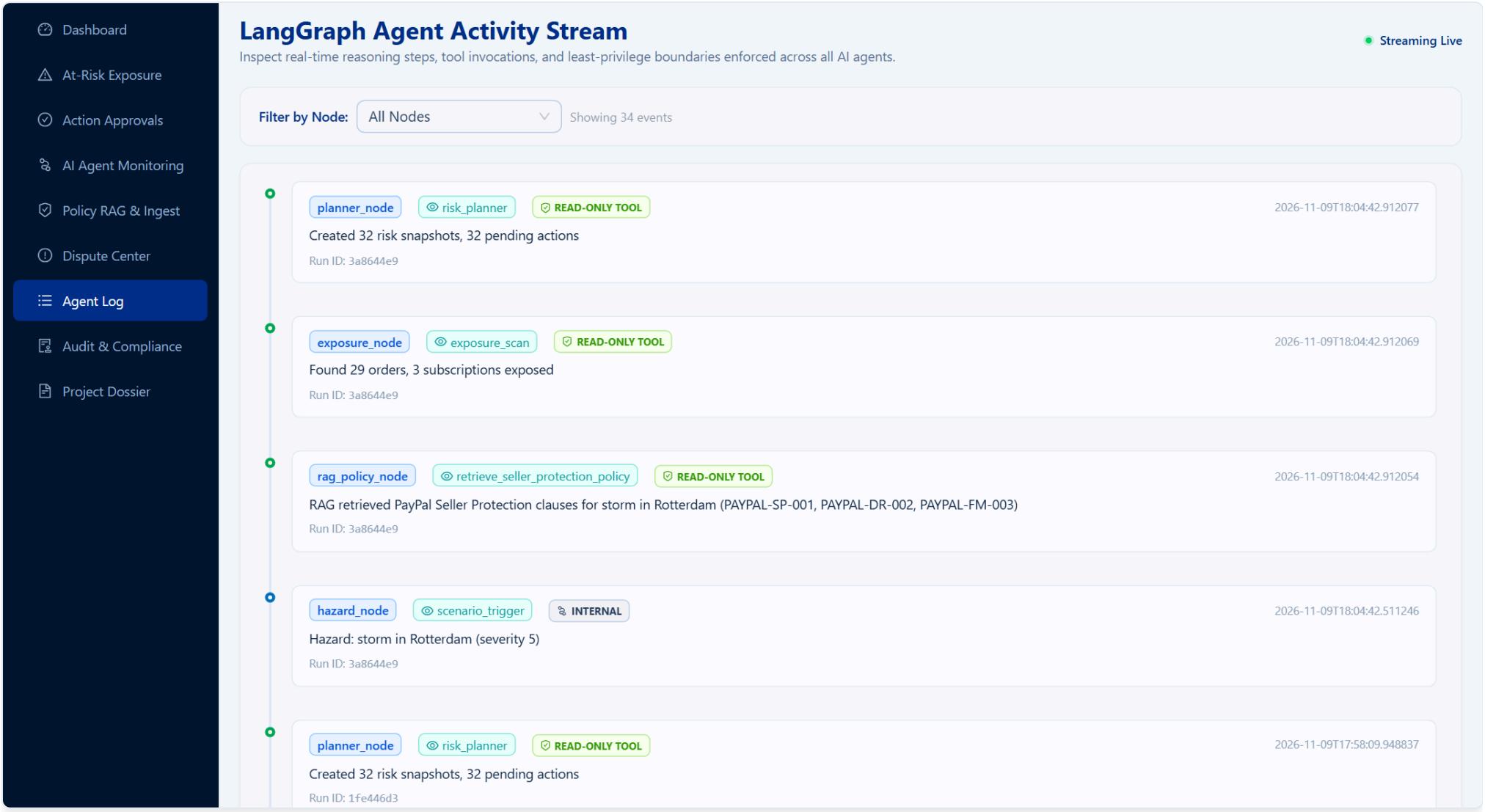
> Real-time stateful graph execution trace visualizing HazardSentinel, RiskPlanner, PolicyGrounder, and mandatory human write gates.



Step 9: Multi-Agent Observability & Tool Isolation

Workflow Step 14 of 15

> Execution audit of LangGraph hazard sentinels, verifying strict read-only tool confinement and deterministic human escalation boundaries.



Step 10: Compliance & Immutable Audit Ledger

Workflow Step 15 of 15

Cryptographic ledger recording all logins, UUID idempotency tokens (PayPal-Request-Id), and human approval events for compliance auditability.

